EasyA Competition: Result Analysis

Ryan Lam

February 12, 2022

Introduction

The EasyA Math challenge has come to a close with 15 submitted answer scripts. Congratulations to everyone who had worked so hard! As the lead marker of the competition, I would like to thank Phil, Karanveer and Krish for helping me to communicate and promote the competition, Nilo and Alex for marking the papers and proofreading them, and finally Edwin and Jonas for writing up the solution for some of the question!

The questions and their suggested answer, and more info about the competition can be found on my website here.

In this section, we try to present some key statistics after marking the results, and some very nice correlations between the section marks and total marks!

Summary Statistics

Each team's attempt consist of their Part A, B, and C marks and their total marks. Since Part B and C consisted of the harder questions, I combine the marks earned for two part into a variable `PartBC`.

```
#import data
data = read.csv("EasyA.csv")
cat("The Mean and Standard Deviation for Part A is",
    mean(data$PartA), " and",
    sd(data$PartA), "marks \n")
```

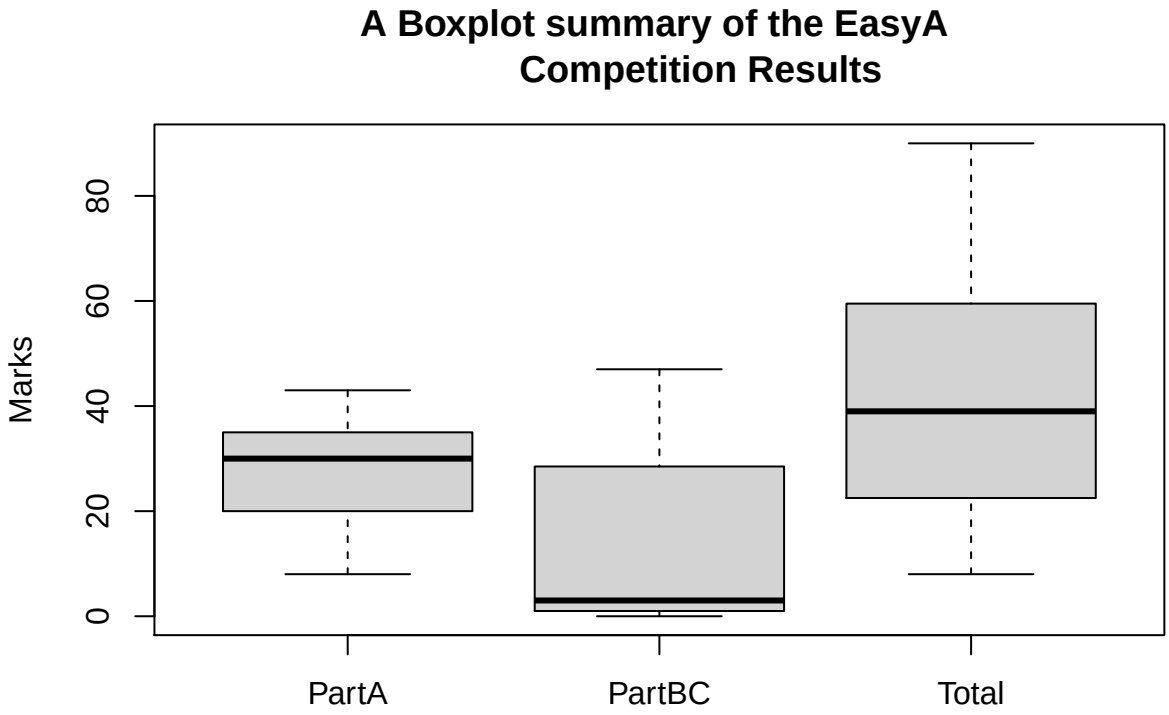
```
## The Mean and Standard Deviation for Part A is 27.2  and 10.16436 marks
```

```
cat("The Mean and Standard Deviation for Part B&C is",
    mean(data$PartBC), " and",
    sd(data$PartBC),"marks \n")
```

```
## The Mean and Standard Deviation for Part B&C is 15.6  and 16.89379 marks
```

We could also use other visualizations:

```
#Boxplot plotting command
boxplot(data, ylab = "Marks",
        main = "A Boxplot summary of the EasyA
        Competition Results")
```



Linear Regression On Results

There is also something that can be reported by a linear regression model with total marks against marks in Part B and C (albeit not a very good statistical practice because we know they are correlated by the formula `Total = PartA + PartBC`, but we are going to try...)

```
#Linear Model
m1 = lm(data = data, Total ~ PartBC)
summary(m1)
```

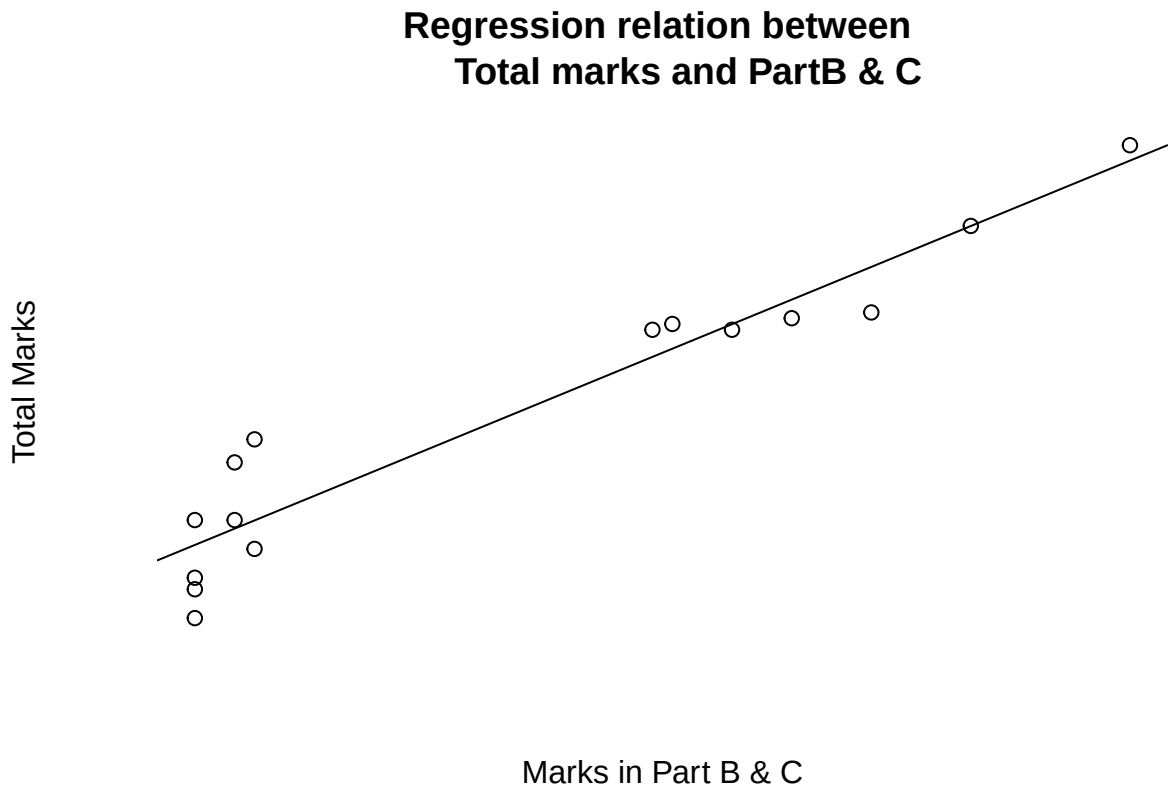
```
##
## Call:
## lm(formula = Total ~ PartBC, data = data)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -12.6790  -5.3061   0.0186   4.3048  14.0669
##
## Coefficients:
##             Estimate Std. Error t value Pr(>|t|)
## (Intercept)   20.679      2.710   7.631 3.73e-06 ***
## PartBC         1.418      0.120  11.815 2.52e-08 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 7.587 on 13 degrees of freedom
## Multiple R-squared:  0.9148, Adjusted R-squared:  0.9082
```

```
## F-statistic: 139.6 on 1 and 13 DF,  p-value: 2.518e-08
```

A very high R^2 value (≈ 0.91) might mean that it is okay to predict the Total marks from marks in Part B and Part C combined.

In other words, Part B and C really differentiate the performance of this paper! Let's see this graphically.

```
#axes removed to prevent Teams
#knowing each other's actualy marks ;)
plot(data$PartBC, data$Total,
     main = "Regression relation between
     Total marks and PartB & C",
     xlab = "Marks in Part B & C",
     ylab = "Total Marks", axes = FALSE)
abline(m1$coefficients)
```



Final Comments

As a lead marker and question setter of the competition, I am very happy on how this competition ends on such a high note. Hopefully we can see more of you joining us in the future!